

See Chaitin's proof

#### Chaitin's proof of Gödel's theorem

Statements such as  $K(n) > m$   
cannot be proven above a certain value of  $m$ ,  
though they are true for infinitely many integers  $n$ .

Suppose that for infinitely many  $m$ , there is an integer  $x_m$  such that  $K(x_m) > m$  is provable. The value of  $x_m$  can be determined by an algorithm that enumerates all theorems. This algorithm has only  $m$  as input. Therefore:  $K(x_m) < 2 \log(m) + O(1)$  (using a doubling code as prefix code for  $m$ ). We get:  $m < K(x_m) < 2 \log(m) + O(1)$ , which is absurd as soon as  $m$  is large enough.

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